EarlyCare AI

An AI agent for early detection of chronic diseases using symptom checks and machine learning



SDG 3: Good Health and Well-being | IBM SkillsBuild Masterclass 5 Project

Agenda

01 SDG alignment & problem statement

02 The problem in numbers

03 Existing alternatives & gaps

04 Our solution & how it works

05 Architecture, AI models & tech stack

06 Users, business model & Lean Canvas

07 Impact, ethics & roadmap

Aligned with SDG 3



SDG 3

Ensure healthy lives and promote well-being for all at all ages.

Target 3.4

Reduce premature deaths from non-communicable diseases by one third through prevention and treatment.

Target 3.8

Achieve universal health coverage, including access to quality essential health services.

Target 3.c

Increase health financing and the recruitment, training and retention of the health workforce.

Problem statement

> *Millions of Indians live with diabetes, heart disease, hypertension and kidney disease without knowing it, because screening is irregular and doctors are far away. This project builds an AI agent that uses a symptom checker and machine learning on basic health data to flag risk early and guide people to the right care.*

Who

Adults 30+ in Tier-2/3 towns & villages

When

Before symptoms turn into complications

Goal

Earlier testing, earlier treatment

The problem in numbers

~2 in 3

deaths in India are caused by non-communicable diseases

101 M

Indians live with diabetes

136 M

more have prediabetes, a window where disease can be prevented

~80%

shortfall of specialists at rural community health centres

Sources: World Bank (NCD deaths, 2019: 65.9%); ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023; Health Dynamics of India 2022-23, MoHFW.

Why disease is detected too late



Silent symptoms

Diabetes, BP and kidney disease show few signs until damage is done.



Distance & cost

Specialists and labs sit in cities; a check-up can cost a day's wages.



Language & literacy

Most health apps are English-only and assume medical knowledge.



No follow-up

Camps screen once; nobody reminds people to test again or see a doctor.

Existing alternatives fall short

Alternative	What it offers	Gap
Google / global symptom-checker apps	Instant answers to symptoms	English-first, no Indian context, no follow-up
Paid full-body check-up packages	Accurate lab tests	Costly, city-based, done rarely
Government NCD screening camps	Free screening at community level	Infrequent, limited reach, paper records
Waiting for symptoms	No cost today	Late diagnosis, expensive treatment

EarlyCare AI fills the gap: free, local-language, continuous screening with a clear next step.

Our solution: EarlyCare AI

A health-check camp in your pocket

A free, multilingual AI agent that talks to people in their own language, estimates their risk of chronic disease in about 3 minutes, and tells them exactly what to do next.

Screening and guidance, never a diagnosis.



Talk

Chat or voice in Kannada, Hindi or English



Assess

ML models score risk for 4 chronic diseases



Act

Nearest PHC or lab, teleconsult, reminders

How it works: the user journey



1. Start

User opens WhatsApp bot or app, picks a language



2. Answer

Agent asks about symptoms, lifestyle & family history



3. Measure

Optional: BP, sugar, weight from camp or home device



4. Predict

ML models return a risk level with reasons



5. Next step

Test, doctor visit or healthy habits + reminders

Results are saved so the agent can check in again in 3 to 6 months and track change over time.

Key features

Local-language voice & chat

Kannada, Hindi, English to start; voice for low-literacy users

Multi-disease risk scoring

Diabetes, heart disease, hypertension, chronic kidney disease

Explainable results

Shows top factors behind each risk, e.g. BMI or family history

Care finder

Nearest PHC, government lab or teleconsultation option

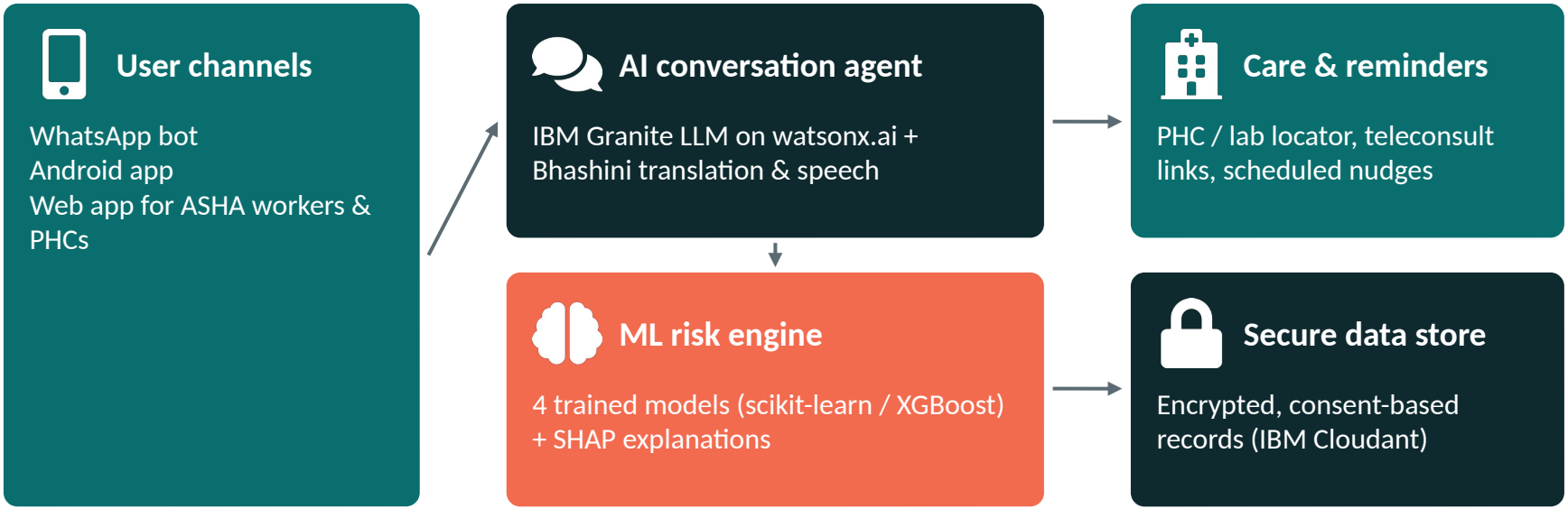
Follow-up reminders

Nudges to get tested, take medicine, re-screen

ASHA worker mode

Screen a whole household and share a summary with the PHC

System architecture



Backend: Python FastAPI on IBM Cloud Code Engine. All services talk over HTTPS with role-based access.

How the AI finds risk

1. Learn

Models learn from public health datasets of past patients

2. Check

Your answers are compared with patterns the models learned

3. Score

You get a low, medium or high risk level, with the reasons

What it checks for

- Diabetes
- Heart disease
- High BP
- Kidney disease

Goal: catch at least 85 of every 100 people who are truly at risk.

Technology stack



AI & LLM

IBM watsonx.ai, IBM Granite models, prompt templates for safe triage



Machine learning

Python, pandas, scikit-learn, XGBoost, SHAP



Language & voice

Bhashini APIs for Indian-language translation, speech-to-text



Backend

FastAPI, REST APIs, IBM Cloud Code Engine



Front end

WhatsApp Business API, Android app, Streamlit dashboard



Data & security

IBM Cloudant, encryption at rest, consent logs

Who we serve

Individuals

Adults 30+ in Tier-2/3 towns and villages

ASHA & ANM workers

Frontline workers who screen households

PHCs & health missions

Public health centres and state programmes

Employers & insurers

Wellness programmes that want healthier members

Early adopters

- Smartphone users aged 30-60 who already worry about sugar or BP
- ASHA workers in one pilot district in Karnataka
- NGOs and college NSS units already running free health camps

Lean Canvas at a glance

Problem

Late detection of chronic disease; unscreened prediabetes; specialist shortage

Solution

Multilingual AI agent + ML risk scores + clear next step

Unique value

Know your risk in 3 minutes, in your language, for free

Unfair advantage

Rural-first design, PHC/ASHA partnerships, India-specific data

Customers

Adults 30+ in Tier-2/3 & rural areas; ASHAs; PHCs; employers

Channels

ASHA workers, health camps, WhatsApp, Play Store

Key metrics

Screenings, follow-up test rate, model recall

Costs

Cloud & AI, development, clinical review, outreach

Revenue

Free for users; B2G & B2B licences; lab referrals; CSR

Full canvas: EarlyCare_AI_Lean_Canvas.pdf

Business model & sustainability

Free

Core screening is always free for individuals. Others pay for scale, data insights and integration.

B2G licences

Per-screening fee or annual licence for state health missions & PHCs

B2B wellness plans

Per-employee pricing for employers and insurers

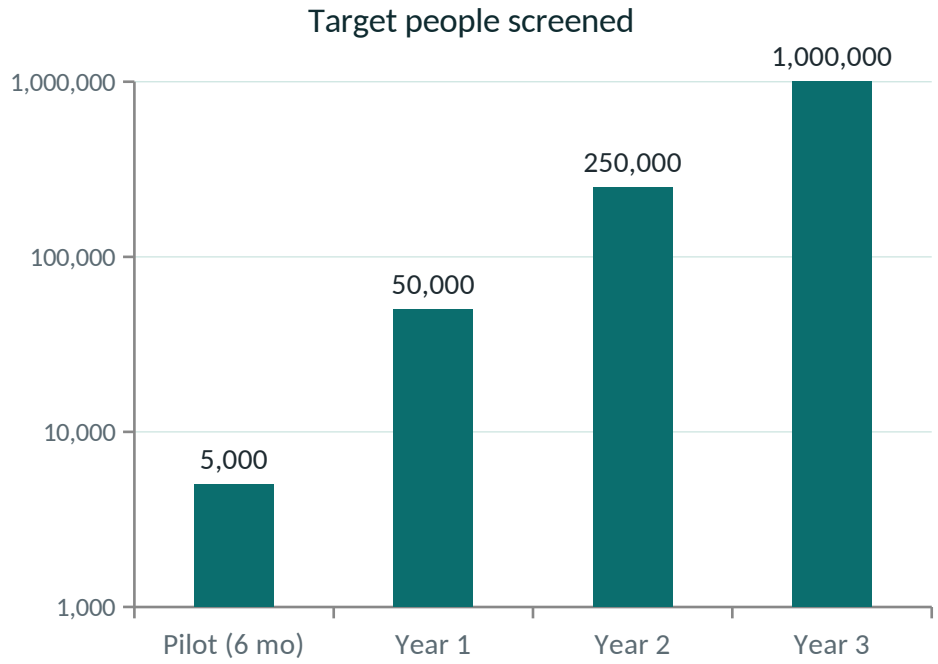
Lab referrals

Fee from partner labs for confirmatory tests booked via the app

CSR & grants

Health-tech CSR funds and innovation grants for rural roll-out

Expected impact & key metrics



85%+ model recall on validation data

50% of high-risk users get a confirmatory test in 30 days

< Rs 10 cost per screening

Targets are planning estimates for the pilot; chart uses a log scale.

Responsible AI: privacy, safety & fairness



Not a diagnosis

Clear disclaimers; high-risk results always route to a doctor or PHC



Privacy by design

Consent first, minimal data, encryption; follows India's DPDP Act 2023



Fairness

Test model accuracy across age, gender and rural/urban groups



Human oversight

Clinical advisor reviews questions, thresholds and red-flag rules



Transparency

Every risk score shows the factors that drove it



Inclusion

Voice and local languages for low-literacy and elderly users

Implementation roadmap

Month 1-2

1

Research & data

Study datasets, consult a doctor, design questions in 3 languages

Month 3-4

2

Build MVP

Train & validate models, build chatbot on watsonx.ai, WhatsApp bot

Month 5-6

3

Pilot

Test with ASHA workers and one NGO camp in a Karnataka district

Month 7-12

4

Improve & scale

Add languages and diseases, PHC dashboard, partner onboarding

Thank you

Catch disease early. In every language. For everyone.



EarlyCare AI | SDG 3: Good Health and Well-being | Questions welcome